

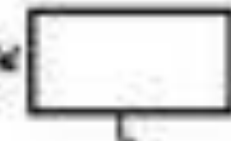
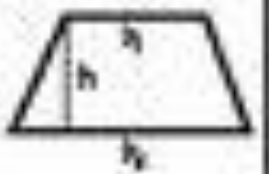
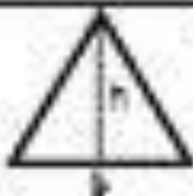


Mensuration

By Aanchal Kansal

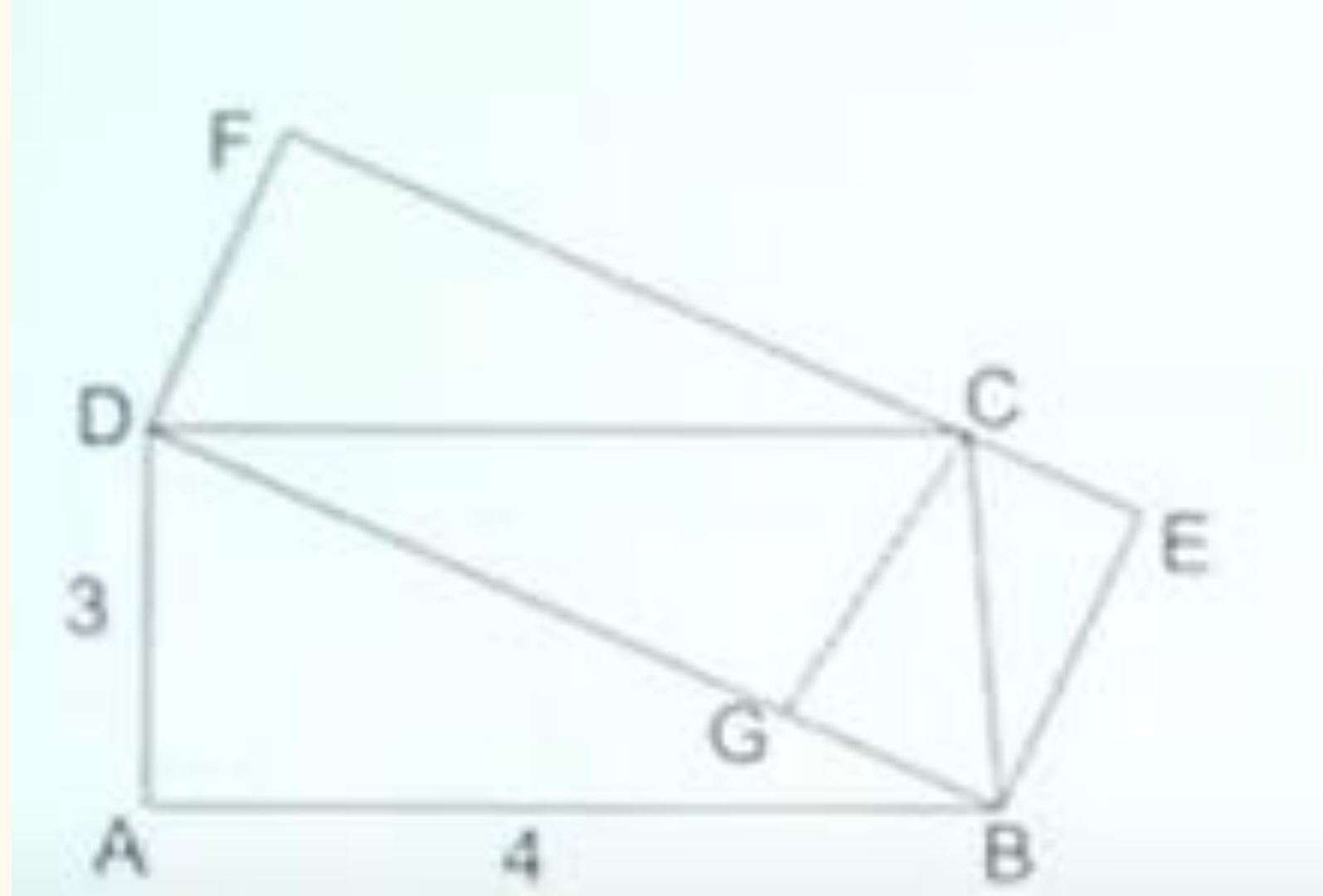
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Area & Perimeter Formulas

Shape	Area	Perimeter
Circle 	$A = \pi r^2$	Circumference $C = \pi d = 2\pi r$
Parallelogram 	$A = b \times h$	$P = 2b + 2s$
Rectangle 	$A = L \times W$	$P = 2L + 2W$
Rhombus 	$A = \frac{(d_1 d_2)}{2}$	$P = 4s$
Square 	$A = s^2$	$P = 4s$
Trapezoid 	$A = \frac{(b_1 + b_2)h}{2}$	$P = b_1 + b_2 + s_1 + s_2$
Triangle 	$A = \frac{bh}{2}$	$P = s_1 + s_2 + s_3$

Two rectangles ABCD and DBEF are as shown in the figure.
The area of rectangle DBEF (in square units) is

- A. 10
- B. 12
- C. 14
- D. 15



The ratio of areas of two squares, if diagonal of one is double the diagonal of the other is

- A. 3 : 2
- B. 4 : 1
- C. 3 : 1
- D. 4 : 3

The area of a square is 484 cm^2 . What is its perimeter?

- A. 88 cm B. 66 cm C. 44 cm D. 22 cm

If the radius of a circle is doubled, its area becomes:

- A. 2 times B. 4 times C. 8 times D. Half

A rectangle has dimensions 18 cm $\times$ 12 cm. If each side is increased by 50%, the new area is:

- A. 486 cm² B. 972 cm² C. 648 cm² D. 1296 cm²

A circular garden has a circumference of 44 m. Find the radius and area of the garden

A square and a rectangle have the same area of 144 cm^2 . The length of the rectangle is 18 cm. What is its breadth? Which shape has the greater perimeter

If the diagonal of a square is $10\sqrt{2} \text{ cm}$, what is its area?

A. 100 cm^2 B. 50 cm^2 C. 200 cm^2 D. 150 cm^2

A square and a rectangle have the same perimeter of 64 cm. If the rectangle's length is 20 cm, what is its area?

- A. 192 cm^2 B. 220 cm^2 C. 176 cm^2 D. 144 cm^2

The diagonals of a rhombus are 16 cm and 12 cm. Find its area.

A square and a circle have the same area. If the side of the square is 14 cm, what is the radius of the circle?

A path 1.5 m wide is built inside a rectangular garden 20 m by 12 m. Find the area of the path

A semi-circular window has a diameter of 2.8 m. What is its area?

A wire is bent into the shape of a square. If its length is 88 cm, find the area it encloses.

A rectangle and a triangle have the same base of 10 cm and the same area of 60 cm^2 .
What is the height of the triangle?